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Linear equations in one unknown: Construction

Contents and concept route



An equation constrains an unknown

Subject: Mathematics

Topic: Linear equations

Concept: An equation constrains an unknown

Concept difficulty: Medium concept

1

Test a value by putting it into the equation and comparing the two sides, before changing the equation at all.

Medium concept

Section 1 of 3

Next →

Conditions for this construction

- Can evaluate an arithmetic expression containing a given number.

Prerequisite reference

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1 Declare the domain the unknown ranges over.

Why valid: A solution set has no meaning until the admissible values are fixed.

Result: x ranges over the rationals.

2 Substitute a candidate value and evaluate both sides separately.

Why valid: Membership of the solution set is decided by substitution, which is the definition of a solution.

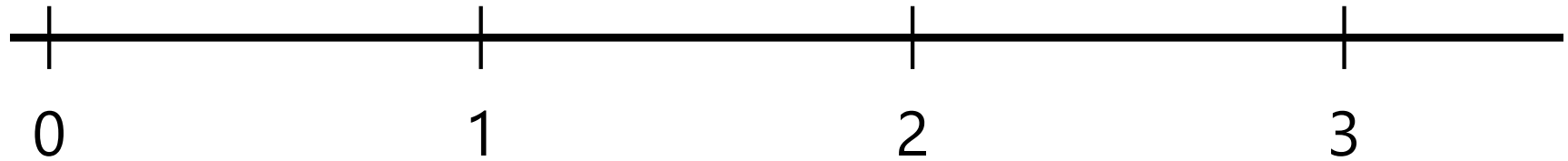
Result: For $x = 2$: $3(2) + 2 = 8$, and the right side is 9, so $8 = 9$ is false.

3 Try a second value that fails, and say why it fails.

Why valid: A constraint is only understood once a non-solution has been seen.

Result: Evidence that the equation selects values rather than computing one.

Domain: the rationals · x (dimensionless)



Draw the rationals as one labelled axis with integer ticks before anything is marked.

Stage 1 of 3

Place the exact solution between its bracketing integers so that it is visibly not equal to either.

The key step

Read $a \cdot x + b = c$ as a statement that is true for some values of the unknown and false for others, rather than as an instruction to work out a left-hand side.

Worked explanation

The unknown x ranges over the rationals. Test $x = 3$ and $x = 4$ in $2x + 5 = 11$. Which of them, if either, is a solution?

Follow the operation, its justification and its result.

1. Both 3 and 4 are rationals, so both are admissible candidates.
2. For $x = 3$ the left side is $2 \times 3 + 5 = 11$, the right side is 11, and $11 = 11$ is true.
3. For $x = 4$ the left side is $2 \times 4 + 5 = 13$, and $13 = 11$ is false: it overshoots by 2.

Result

$x = 3$ is a solution; $x = 4$ is not.

Check

A value in between, $x = 3.5$, gives $2 \times 3.5 + 5 = 12$, which is also not 11: the equation keeps selecting values, and each increase of 1 in x raises the left side by 2.

Mistake clinic • diagnose and repair



1-2-3 QUICK CHECK

1 • CHECK

Evaluate each side on its own for the candidate: a solution makes the two numbers identical, not merely close.

2 • CHECK

Confirm the candidate belongs to the declared domain before testing it; a value outside the domain is not a solution even if the sides agree.

Now you do one

Does $x = 2$ satisfy $3x + 2 = 9$? Show how you decided.

Operations that preserve the solution set

Subject: Mathematics

Topic: Linear equations

Concept: Operations that preserve the solution set

Concept difficulty: Medium concept

1

Choose each step so that it can be undone, and say why, including that the number you divide by is not zero.

Medium concept

Section 2 of 3

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Conditions for this construction

- Can decide by substitution whether a value satisfies an equation.

Prerequisite reference

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1 Subtract b from both sides.

Why valid: Adding the same quantity to both sides can be undone, so the solution set is unchanged.

Result: $a \cdot x = c - b$

2 State that a is non-zero, then divide both sides by a.

Why valid: Dividing by a non-zero quantity is reversible; division by zero is undefined, so the condition must be stated rather than assumed.

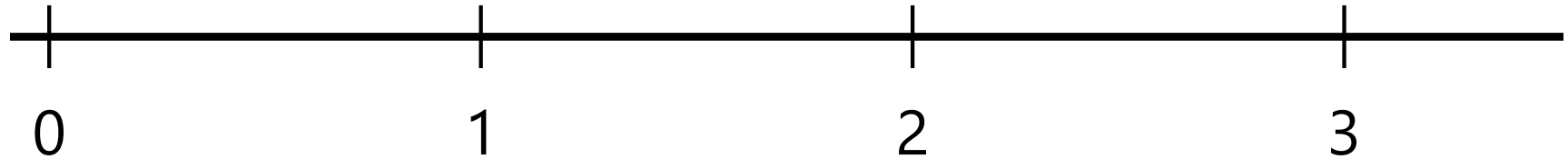
Result: $x = (c - b)/a$

3 **Substitute the result into the original statement, not into an intermediate line.**

Why valid: Checking against the original catches an error introduced at any step.

Result: A verified solution.

Domain: the rationals · x (dimensionless)



Draw the rationals as one labelled axis with integer ticks before anything is marked.

Stage 1 of 3

Place the exact solution between its bracketing integers so that it is visibly not equal to either.

The key step

See each manipulation as an operation applied to both sides whose reversibility is what guarantees no solution is gained or lost.

Formula reference · meaning and conditions

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Worked explanation

Solve $3x + 2 = 9$ over the rationals. Give the exact solution and verify it.

Follow the operation, its justification and its result.

1. **Read $3x + 2 = 9$ as a claim about x : x is first multiplied by 3, then 2 is added.**

Why valid: Undoing needs the order in which operations were applied to the unknown.

Result: x is tripled, then 2 is added.

2. **Undo the last operation first: subtract 2 from both sides.**

Why valid: Subtracting the same number from both sides is reversible, so the solution set is unchanged.

Result: $3x = 7$

3. **Divide both sides by 3.**

Why valid: 3 is non-zero, so dividing by it is reversible.

Result: $x = 7/3$

4. **Substitute $x = 7/3$ into the original equation.**

Why valid: A solution must make the original claim true exactly.

Result: $3 \times 7/3 + 2 = 7 + 2 = 9$, as required.

Result

$x = 7/3$.

Check

The truncation 2.33333333333 gives 8.99999999999, not 9, so the exact fraction is the solution and the decimal is not.

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1-2-3 QUICK CHECK

1 · CHECK

Run the steps backwards from the answer: multiply by a , then add b . You must land on c ; if a step cannot be undone, the solution set may have changed.

2 • CHECK

Put the answer into the original equation, not into a line you wrote along the way, so an error at any step shows up.

Now you do one

Solve $5x - 4 = 11$, naming the operation used at each step and why it is reversible.

An exact solution is not its decimal approximation

Subject: Mathematics

Topic: Linear equations

Concept: An exact solution is not its decimal approximation

Concept difficulty: Medium concept

1

Keep the solution as a fraction in lowest terms, and prove by substitution that a cut-off decimal is a different number.

Medium concept

Conditions for this construction

- Can isolate the unknown using reversible operations.

Prerequisite reference



1 Record the solution as a fraction in lowest terms.

Why valid: The division producing it need not terminate as a decimal, so the fraction is the exact value.

Result: $x = 7/3$

2 Substitute the exact fraction into the original statement.

Why valid: Exact substitution reproduces the statement with no residue.

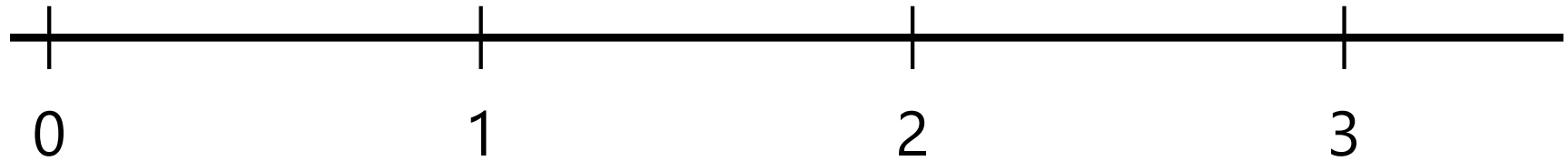
Result: 3 times $7/3$, plus 2, is exactly 9.

3 Substitute a truncated decimal and compare.

Why valid: Seeing the residue is what distinguishes the two objects; being told they differ is not the same as observing it.

Result: 3 times 2.333333333333, plus 2, is 8.999999999999.

Domain: the rationals $\cdot x$ (dimensionless)



Draw the rationals as one labelled axis with integer ticks before anything is marked.

Stage 1 of 3

Place the exact solution between its bracketing integers so that it is visibly not equal to either.

The key step

Treat the exact rational and its truncated decimal as two different numbers, and use substitution to show the difference rather than asserting it.

Worked explanation

Solve $6x - 1 = 3$ over the rationals. Give the exact solution, then decide whether 0.666 also satisfies the equation.

Follow the operation, its justification and its result.

1. Add 1 to both sides: $6x = 4$.
2. Divide both sides by 6, which is not zero: $x = 4/6$, which is $2/3$ in lowest terms.
3. Substitute $2/3$: $6 \times 2/3 = 4$, and $4 - 1 = 3$, exactly the right side.
4. Substitute 0.666: $6 \times 0.666 = 3.996$, and $3.996 - 1 = 2.996$, which falls short of 3 by 0.004.

Result

$x = 2/3$; 0.666 is not a solution, because it makes the left side 2.996.

Check

The denominator of $\frac{2}{3}$ is 3, which has a prime factor other than 2 or 5, so no terminating decimal can equal it; 0.666 had to fail.

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1-2-3 QUICK CHECK

1 • CHECK

Bracket the fraction between whole numbers first: $\frac{7}{3}$ lies between 2 and 3 because $3 \times 2 = 6 < 7 < 9 = 3 \times 3$. A reported value outside that bracket is wrong, whatever its decimals.

2 • CHECK

Look at the denominator in lowest terms: if it has a prime factor other than 2 or 5, as 3 does, no terminating decimal can equal the solution.

Now you do one

Solve $3x + 2 = 9$ over the rationals, and explain why writing the answer as 2.33 would be a different claim.

